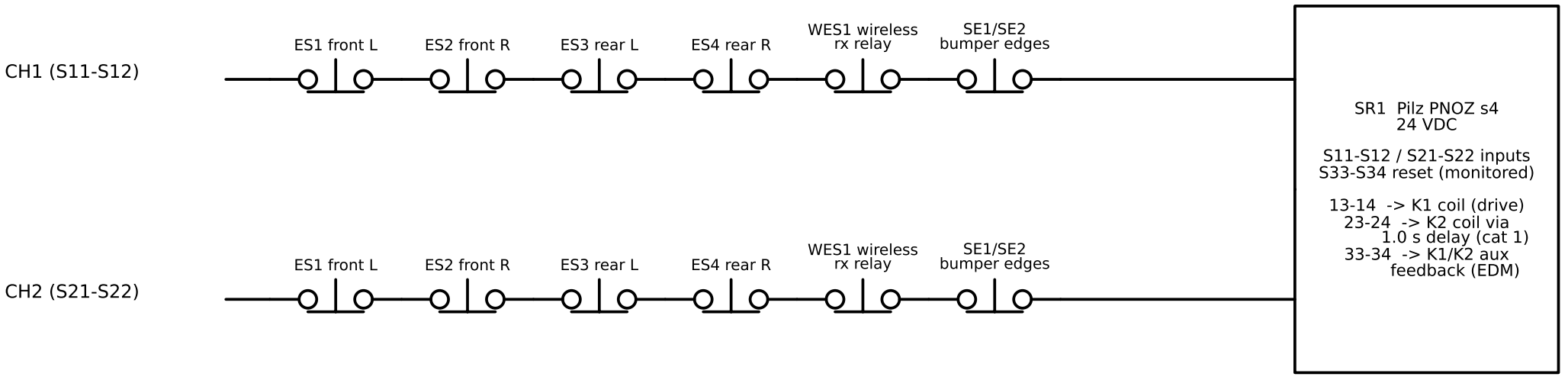
E-02 Safety chain -> PNOZ s4 -> K1 drive / K2 arms

Dual channel, cross-monitored, Cat 3 / PL d target (ISO 13849-1). Stop category 0 for drive, cat 1 for arms (controlled stop then K2).



Also wired into SR1 inputs in series: FR5 controller safety I/O (each arm reports protective stop), BMS fault relay.
Software stop (non-safety) path: Jetson -> Teensy -> VESC zero current + FR5 'stop' over Ethernet; used for normal pauses.
Geofence + speed limit (0.5 m/s near people detected by perception) are software functions and are NOT counted as safety functions in v1.
Production: add SICK outdoorScan3 safety laser scanner zones (protective field 1.5 m at 1.5 m/s) to reach ISO 18497 people-detection.